MAIS PAC

IPhO 02.06: Electrostatics and Steady Currents · Worked Solutions

Electrostatics and Steady Currents: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Three charges in a line. Return →

(a) F = kq_1q_2/r² = (8.99×10^9)(2.0×10^(-6))(3.0×10^(-6))/(1.2)² = 0.037 {N}, repulsive.

(b) Let the third charge sit at distance x from +2.0 μ{C} toward +3.0 μ{C}. For zero net field the field magnitudes must cancel:

$$(2)/(x^2) = (3)/((1.2-x)^2)\ (1.2-x)/(x) = \sqrt{(1.5)} = 1.225\ x = 0.539\ \{m\}. \qquad (1)$$

The other root lies outside the segment and corresponds to fields pointing the same way.

(c) V = k((2)/(0.539) + (3)/(0.661))×10^(-6) = 8.99×10^9×(3.71+4.54)×10^(-6) = 7.4×10^4 {V}. **Check:** zero **field** does not mean zero **potential** — a distinction that carries marks.

Solution 2 — Field of a charged sphere. Return →

(a) By Gauss, outside E = kQ/r². Inside, the enclosed charge is Qr³/R³, so E = kQr/R³.

(b) Outside, V = kQ/r. Inside, integrate inward from the surface with the field kQr/R³: V(r) = (kQ)/(2R)(3 - (r²)/(R²)).

(c) E drops to zero at the surface only if the charge is a shell; for a solid sphere E is continuous but its derivative jumps, because the charge density is discontinuous there. V is continuous always — otherwise the field (its gradient) would be infinite. **Check:** at r=R both formulas give kQ/R for V and kQ/R² for E, as continuity requires.

Solution 3 — Capacitors in series and parallel. Return →

(a) Parallel: C = 11 μ{F}, U = (1)/(2)(11×10^(-6))(144) = 7.9×10^(-4) {J}.

(b) Series: (1)/(C) = (1)/(2)+(1)/(3)+(1)/(6) = 1, so C = 1.0 μ{F} and U = (1)/(2)(10^(-6))(144) = 7.2×10^(-5) {J}.

(c) At a fixed supply voltage, energy is (1)/(2)CV², so it scales directly with the capacitance. In series the total voltage is shared and the equivalent capacitance is smaller than the smallest individual one. **Check:** 11 times less energy for exactly 11 times less capacitance — the ratio is consistent.

Solution 4 — Charging a capacitor. Return →

(a) ε = IR + q/C with I = dq/dt gives R q + q/C = ε, whose solution with q(0)=0 is q = Cε(1-e^(-t/RC)).

(b) The battery supplies ∫ε I,dt = ε Q = Cε². The capacitor stores (1)/(2)Cε². The remaining half is dissipated in the resistor as heat, whatever the value of R — a result that surprises everyone once.

(c) At t = RC, q/Q = 1 - e^(-1) = 0.632, so 63%. **Check:** as R→0 the **power** diverges as 1/R while the **duration** shrinks as R, so the energy stays (1)/(2)Cε²; the limit is still consistent.

Solution 5 — A Wheatstone bridge. Return →

(a) Both arms divide 6.0 {V} in the ratio 1:2, so the midpoints sit at 4.0 {V} and 4.0 {V}: zero potential difference, so no current.

(b) Balanced when (R_1)/(R_2) = (R_3)/(R_4).

(c) With R_4 = 33 Ω the right midpoint becomes (33)/(48)×6 = 4.125 {V}, above the left midpoint at 4.0 {V}, so current flows from right to left through the galvanometer. **Check:** the reasoning uses only potential division, not the full three-loop solution — the physically correct shortcut.

Solution 6 — Drift, mean free time and the electron gas. Return →

(a) I = nAev_d v_d = (2.0)/((8.5×10^(28))(10^(-6))(1.6×10^(-19))) = 1.5×10^(-4) {m,s^(-1)}.

(b) From ρ = m_e/(ne²τ): τ = (9.11×10^(-31))/((8.5×10^(28))(1.6×10^(-19))²(1.7×10^(-8))) = 2.5×10^(-14) {s}.

(c) Thermal speeds are √(2kT/m)10^5 {m,s^(-1)}, nine orders of magnitude larger than the drift speed. The electrons do not travel slowly; they scatter constantly and drift **collectively** slowly, so a signal travels at nearly the speed of light while individual electrons creep. **Check:** the mean free path vτ2.5 {nm} is a few tens of atomic spacings, as expected for a metal.

Solution 7 — Two spheres connected by a wire. Return →

(a) Equal potentials: kQ_1'/r_1 = kQ_2'/r_2 with Q_1'+Q_2' = Q_1+Q_2, so

$$Q_1' = (Q_1+Q_2)(r_1)/(r_1+r_2),\ Q_2' = (Q_1+Q_2)(r_2)/(r_1+r_2). \qquad (2)$$

(b) σ_1' = Q_1'/(4π r_1²) ∝ (1)/(r_1(r_1+r_2)), and similarly for 2, so σ_1'/σ_2' = r_2/r_1. The fields at the surfaces, E = σ/ε_0, follow the same ratio squared relative to the spheres' own charges.

(c) A sharp point is a small-radius conductor: it carries a high surface charge density and therefore a strong field, which ionises the air and starts a discharge. **Check:** as r_1→ r_2 the charges equalise, as they should for identical spheres.

Solution 8 — Internal resistance and maximum power. Return →

(a) I = 12/5.5 = 2.18 {A}, V_({terminal}) = 12 - 2.18(0.50) = 10.9 {V}.

(b) P = (ε²R)/((R+r)²) is maximal at R = r = 0.50 Ω, giving P = (144)/(4(0.5)) = 72 {W}.

(c) At R=r the load and internal resistance each dissipate 72 W, so the efficiency is 50%. Real supplies are designed for low internal resistance and operation below the maximum-power point, because efficiency matters more than power. **Check:** at R→∞ and R→0 the power delivered tends to zero, so the maximum in between is genuine.

Solution 9 — Potentiometer. Return →

(a) The wire carries 2.0/12 = 0.167 {A}, so the potential gradient is 2.0 {V/m}. A balance length of 0.620 {m} corresponds to ε_x = 1.24 {V}.

(b) At balance, no current flows through the unknown cell, so its terminal voltage equals its emf; the balance length is proportional to ε_x/ε_({driver}) times the length. Only the **uniformity** of the wire enters, so a precise total resistance is unnecessary.

(c) A lower driver emf halves the potential gradient proportionally, so the balance length becomes 0.620×(2.0/1.9) = 0.653 {m}. **Check:** longer wire for the same unknown cell at a lower driver emf — exactly what a smaller gradient implies.

Solution 10 — A sliding contact and a nonlinear element. Return →

(a) While the diode conducts, the loop rule is 9.0 - 0.70 - IR(x) = 0, so

$$I(x) = (8.3)/(6.0(1 + 3x/L)) \text{ {A}}. \tag{3}$$

(b) The diode is forward-biased as long as the open-circuit voltage across its position would exceed 0.70 {V}. Here the source is 9.0 {V} and the only series element is R(x) 6.0 Ω, so the diode never turns off for any finite x: the current merely falls from 1.38 {A} at x=0 to 0.35 {A} at x=L. The physically important statement is that a pure series resistance cannot make a diode stop conducting; you need a second voltage source or a change of bias direction.

(c) Kirchhoff's loop rule is energy conservation per unit charge and holds for any element; the nonlinearity only means the V–I relation is a curve, so the loop equation must be solved **together with** that curve (graphically, or by a load-line construction). **Check:** the current is always positive and decreases monotonically with x, consistent with the increasing resistance.